Research Proposal
MSc Thesis in Data Science and Business Analytics

Geospatial Accessibility and Layer Effectiveness in the Portuguese Three-Tier Gatekeeping Healthcare System: A Descriptive Analysis of Emergency Care Structure

by
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Introduction

Due to structural features of Portugal's healthcare system, many portuguese residents bypass primary care services and seek treatment directly at hospital emergency departments, contributing to persistent congestion. Comprehensive indicators such as emergency department capacity or physician density do not fully explain for this pattern. Instead, differences in institutional triage and gatekeeping mechanisms within the healthcare system are likely to play a decisive role.

Previous studies on healthcare accessibility in Portugal have mainly examined geographic inequalities in access to hospitals and primary care services. Research shows clear regional differences, especially between urban areas and low-density rural regions, where access tends to be more limited (Costa et al., 2020; Lopes et al., 2019). Other studies show the uneven distribution of physicians and the ongoing challenge of ensuring equal access across the country (de Oliveira, Dussault and Craveiro, 2017). While these studies help explain the difference in accessibility to healthcare across different regions of Portugal, they focus less on the practical importance of the three-tiered healthcare system. In particular, there is a lack of research exploring whether these tiers function as gatekeeping mechanisms.

Both Portugal and the Netherlands formally operate under a three-tier healthcare structure. In Portugal, this consists of primary care centres (Centros de Saúde), out-of-hours services (Serviço de Atendimento Permanente, SAP), and hospital emergency departments (ED). In the Netherlands, the system includes general practitioners (GPs), out-of-hours GP cooperatives (Huisartsenpost, HAP), and hospital emergency services (SEH). The Dutch system is characterised by strong and operational gatekeeping: patients are required to contact their GP, or the HAP during evenings and weekends, before being referred to hospital emergency care. Emergency departments are therefore designed as a last resort. In contrast, patients in Portugal are not institutionally required to pass through intermediate tiers before presenting at hospital emergency departments, which may result in EDs being structured as central entry points into the system.

This raises an important question: Does Portugal's three-tier healthcare system effectively operate as a gatekeeping structure?

This thesis conducts a descriptive geospatial analysis of Portugal's three-tier healthcare architecture. By constructing population-weighted accessibility models for each tier, the study assesses whether these layers function as effective sequential gatekeepers to emergency care and positions the findings in comparison with the Netherlands.

Research questions

To what extent does the three-tier structure of the Portuguese healthcare system operate as an effective gatekeeping mechanism for access to emergency departments when analysed through geospatial accessibility analysis?

Methodology and Techniques

This study will employ geospatial network analysis and population-weighted accessibility modelling. Travel times will be computed using OpenStreetMap road network data, and Two-Step Floating Catchment Area (2SFCA) methods will be applied to estimate accessibility for each healthcare tier. Descriptive spatial analysis and sensitivity scenarios (e.g., removal of intermediate layers) will be used to evaluate gatekeeping effectiveness and regional variation in emergency department centrality.

Data and/or expected results

This study will use publicly available data, including population statistics (INE : Instituto Nacional de Estatística), healthcare facility locations (primary care centres, SAP units, and hospitals), emergency utilisation data from SNS monitoring portals, and road network data from OpenStreetMap. I expect to identify regional variation in accessibility across the three tiers and assess whether intermediate layers meaningfully alter access to emergency departments. The results may reveal structural differences in layer effectiveness and heterogeneous patterns of emergency department centrality across Portugal.

Literature

Joosten, A. and Gromicho, J. (2026) Health system design, incentives, emergency care, and structural workload: Comparative research notes on Portugal and the Netherlands. Working paper, Internal Medicine Department, Centro Hospitalar Barreiro-Montijo and Amsterdam Business School, University of Amsterdam.

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Timetable

1st Month

Activity	Date
Discuss Research Proposal/Methodology Discuss Literature Review	1. March 9, 2026 2. March 23, 2026

2nd Month

Activity	Date
Discuss Initial Descriptive Analysis Discuss Sensitivity Analysis	3. April 6, 2026 4. April 20, 2026

3rd Month

Activity	Date
Discuss results and conclusions of study Discuss Thesis Draft and Revision	5. May 4, 2026 6. May 18, 2026

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Master's Thesis Business Analytics ■

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Second marker (suggestion by supervisor)
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